Phenomenological Tm models — performance

10-fold cross-validated, n = 235 interfaces. Family A = occupancy-weighted; Family B = elastic / mechanical-coupling. ΔS_Bond and k_rot in cal/mol·K, λ in nm.

Model	cv RMS (°C)	cv R^2	cv ρ	fit R^2	Fitted parameters
A1 mean (g-weighted ΔS_Bond)	2.08	0.785	0.876	0.793	λ=53, ΔS_Bond=−10.07, base=16.05
A2 N-scaled ΔS_Bond (best)	1.98	0.805	0.884	0.811	λ=139.8, ΔS_Bond=−8.13, base=5.24
A3 g on ΔH only	2.05	0.792	0.886	0.795	λ→∞, ΔS_Bond=−7.27, base=4.23
A4 nearest-neighbor g	2.56	0.676	0.748	0.679	λ→∞, ΔS_Bond=−10.6, base=5.93
A5 per-type ΔS_Bond	2.04	0.793	0.873	0.801	λ=57.5; short=−9.94, long=−9.19; base=11.82
A6 split λ_H/λ_S	2.04	0.793	0.878	0.804	λ_H=484, λ_S=754, ΔS_Bond=−8.1, base=5.52
A7 occ. + rotational	2.00	0.802	0.881	0.813	λ=113, ΔS_Bond=−7.98, base=−5.34, k_rot=−0.86
B1 g=1 (no weighting)	2.56	0.676	0.748	0.679	ΔS_Bond=−10.6, base=5.8
B2 g=1 + rotational	2.20	0.761	0.860	0.775	ΔS_Bond=−9.16, base=48.1, k_rot=3.49
B3 g-weighted + WLC tethering	4.05	0.190	0.547	0.201	λ=73.8, base=48.3
B4 WLC tethering only	5.89	−0.715	0.636	−0.699	λ→∞, base=77.1

Best model: A2 (occupancy-weighted, N-scaled per-bond penalty). The elastic Green's-function (rotational) term is degenerate once g is present (A2→A7, no gain); the physically-derived WLC tethering (B3/B4) performs worst. B3/B4 use the corrected asymmetric 6nt bond (8 nt / 7 nt strands).